

Arvind Sharma

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EDUCATION

SASTRA Deemed to be University

Bachelor of Technology in Computer Science Engineering

Thanjavur, Tamil Nadu

June 2020

- **Cumulative GPA:** 8.5411
- **Awards & Honours:** Dean's Merit List (Top 10% in CS Department) for Academic Year 2016-17, 2017-18, 2018-19;

Institute of Management Technology

Post Graduate Diploma in Management (Major: Marketing)

Ghaziabad, Uttar Pradesh

June 2025 – Present

- **Cumulative GPA:** 8.06

WORK EXPERIENCE (4+ years)

Aditya Birla Fashion and Retail Limited

STRIDE BMT INTERN

Mumbai, Maharashtra

April 2026 – June 2026

- Supply Chain Management, Digital Transformation and TMS unit
 - Worked on the TMS integration with ABFRL systems and workflows. Worked closely with the Transportation division of ABFRL SCM team and the 3rd party TMS platform provider team to enable complete coverage across workflows and processes.
 - Improved the coverage of Inbound movements to 97% and Outbound movements to 98%. Regarding the Freight calculation, improved accuracy by 60% for Inbound movements and by 60% for Outbound movements.
 - Worked with cross-functional stakeholders and addressed their operational issues using the TMS.

Tata Consultancy Services

Machine Learning Engineer/ Lead Developer

Chennai, Tamil Nadu

October 2020 – June 2025

- CTO Business Unit(August 2022 – June 2025)
 - Machine Learning Engineer and Technical Lead, handled requirement analysis, design, and delivery of MVPs for a transformative AI-powered product. Expertise in scaling LLMs on multi-GPU (A100/H100) environment, fine-tuning and aligning LLM models on domain data, creating LLM-powered Agents and complex pipelines using prompt engineering.
 - Successful Pilot deployment of AI-powered product within a customer-specific environment onsite(UK)
 - Consulting, Ideating, and Building Generative AI solutions, pipelines, and use cases, as well as proofs-of-concept (POCs) for various Business domains and verticals.
 - Use cases include fine-tuning and aligning open-source models on custom datasets, designing and leading a team to create a workflow using OpenAI and a fine-tuned Stable Diffusion XL model with a refiner, and Domain pipeline automation using GPT.
 - POCs include Talking Avatars, Sales Bot, and Automation using LLM-powered Agents, Prompt Engineering with Open Source model on local consumer machines and consumer GPUs.
 - Worked with AI open-source projects like Auto GPT, Stable Diffusion Web UI, and llama.cpp, vLLM, crewAI, Langchain, TRL, etc.
 - Market Research and Data Analysis on business-related data to enable strategic business choices on emerging technology.
 - Studying and presenting new Generative AI research and Solutions to multiple stakeholders. I have written multiple whitepapers and submitted them to internal forums.
- Corporate Labs (STIL & CMI Rapid Labs), TCS (July 2021 – July 2022)
 - Developed and documented Sales-based use cases using NLP techniques as part of the Sales and Technology Innovation Lab (STIL)
 - In Rapid Labs, developed rapid state-of-the-art Artificial Intelligence solutions (NLP, Image Processing/Computer Vision, Graph NN) as part of POCs and use cases for various Clients across the Communication Media and Information Unit of TCS.
 - POCs include Speaker Diarization, Automated Image Annotation using Vision API, and Q&A tasks using Transformers.
 - Gained competency in 5G core technology and worked on the orchestration of application containers using Kubernetes(EKS).
 - Use cases include the Edge Video Analytics application, an Application platform for an ecosystem-based solution. I took responsibility for Development, Infrastructure Automation using CloudFormation and Ansible, Container Orchestration using Kubernetes (EKS), and Network Mesh using Istio for the use cases mentioned.
 - Studying and presenting multiple emerging technologies across domains and open source libraries to multiple stakeholders.
- Telefonica, UK (December 2020 – June 2021)
 - Took responsibility for monitoring Big Data modules, which provide key insights about the customers.
 - Part of the production upgrade Team, took responsibility for the Analytics application's compatibility with the upgraded environment. Achieved zero data loss.

▪ Acquisition of Hyperspectral Images and Pre-Processing

- Collection and processing of multiple bands belonging to different spectral intensities to form an image band cube.

▪ Semantic Segmentation Network for Classification

- Trained a U-Net model with a band cube of multi- and hyperspectral images and their corresponding annotated images as inputs.

BroadSoft

Developer Intern

Chennai, Tamil Nadu

November 2017 – December 2017

- Got hands-on experience with Software Development Life Cycle. Worked under the development team.

TECHNICAL SKILLS

Languages: Python, Java, YAML, C++**Machine Learning Packages:** NumPy, Pandas, scikit-learn, Tensorflow, Pytorch, keras, transformers, trl, unsloth, accelerate, ray, deepspeed, FSDP, flash-attn, LLaMA-factory, axolotl, langchain, llama-index, llama.cpp, vllm, ollama, lorax, DSpy, whisper.cpp, crewAI, guidance, opencv, diffusers, ultralytics, sentence-transformers, spaCY, h2o, bentoML, dask, MLflow, Kubeflow, faiss, qdrant, haystack, chroma, milvus**Frameworks:** Docker, Kubernetes, Istio, FastAPI, Springboot, PostgreSQL, Git, Helm Charts, Ansible, Redis, Terraform, Kafka, RabbitMQ, celery, gradio, streamlit, ReactJS**Cloud Platform:** AWS (EKS, EC2, Cloud Formation, ECR, IAM, S3), GCP (GKE, Buckets, VM, Cloud Vision API), Azure (AKS, VMs, Disk, OpenAI)**Developer Tools:** Chat-GPT, MS-Office Tools, Adobe Express, VSCode, WSL, Anaconda, GitLab/Github, Swagger editor, Postman**PROJECTS**

▪ **Clinical Trial LLMs**

- Fine-tuned small language models (Llama3, Phi 3 mini, Qwen 2.5 3B) on consumer GPUs using a custom dataset assembled for clinical trials. Used deepspeed and data parallelism to speed up the process on a multi-GPU setup.
- Aligned the fine-tuned models using techniques like ORPO and DPO using a custom dataset collected from fine-tuned models.
- Published some models on the Huggingface [repo](#) under an open-source license.

▪ **Parallel LLM Inference Pipeline**

- Set up multiple scripts using different techniques like batching, parallel inference, etc, for fast LLM inference on a multi-GPU environment.

▪ **Omni-Model**

- A small project demonstrating multi-modal capability on a CPU machine using an ensemble of models, namely whisper, x-gen-mm-phi3, and Nemo TTS model. The flow will take in a video as input and provide audio as output.

▪ **Automatic Image Masking**

- Automatic Image mask generation using a pipeline utilising SAM and Dino model. The entire process was demonstrated on a laptop with an Nvidia GTX 1660 Ti 8 GB graphics card.

▪ **Executive Summary**

- Created a pipeline that generated an executive summary of a document. Used Falcon 40 B model and Flan-T5 model for generating the summary. Utilised a custom Map-Reduce and Refiner setup to achieve a coherent summary.

▪ **Appointment Scheduler**

- Created an end-to-end application for scheduling appointments. Used Springboot for the server and ReactJS for the frontend.

ADDITIONAL

- A keen writer having published technical and business articles in both private and public forums^{[1](#), [2](#)}
- Cricket player who has represented school teams and club teams in the State-level tournament. A football player who has represented the school in the Inter-School tournament. Follow Badminton, Tennis, Mixed Martial Arts, and Motor Sports closely.
- Hobbies include Sketching, Movies and Anime, Photography, Music, and Books.